

UCS503P - Software Engineering Project

Cardio AI: Explainable Heart Disease Prediction

Using CNN-LSTM Attention Network with Grad-CAM Interpretability

Submitted by:

1024170434 Dhruv Singla

1024170451 Srishti

Group: BE Third Year, CSE

Submitted to:

Nisha Thakur

Mid-Semester Evaluation

Computer Science and Engineering Department

Thapar Institute of Engineering and Technology, Patiala

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Chapter 1

Introduction

1.1 Background and Context

Cardiovascular disease remains the leading cause of mortality globally, necessitating advanced, early-prediction systems. While deep learning models can predict heart disease from routine clinical measurements—such as blood pressure, cholesterol levels, and ECG markers—their inherent “black box” nature poses a significant barrier to clinical adoption. Regulatory bodies and medical ethics demand interpretable, auditable predictions. This project, **Cardio AI**, aims to bridge this gap by prioritizing clinical transparency. Our objective is to translate complex deep neural network confidence scores into actionable, human-readable clinical narratives, empowering physicians to trust AI-assisted diagnoses.

1.1.1 Problem Statement

While standard classification tasks can achieve high accuracy on clinical datasets, the primary challenge addressed by this project is shifting the focus from simple heart disease classification to **clinical transparency**—making deep learning explainable rather than a black box. Current challenges in AI-driven heart disease prediction include:

- **Black-Box Predictions:** Deep learning models, while highly accurate, offer no clinical reasoning or transparent decision pathways, leading to physician reluctance.
- **Feature Opacity:** The cardiovascular domain involves heterogeneous clinical features (e.g., numerical vitals, categorical ECG types, binary indicators). Understanding which specific physiological parameters drive a patient’s risk score is non-trivial.
- **Operational Constraints & Overfitting Risks:** Many theoretical models fail in practice due to overfitting on limited datasets (such as the UCI Cleveland dataset). Robust systems must deliver immediate value and incorporate strategies to fail fast through rapid iteration.
- **Lack of Visual Evidence:** Clinicians lack interactive visual artifacts (such as heatmaps or feature importance charts) that map model confidence directly back to patient physiology.

- **Absence of Longitudinal Tracking:** Existing diagnostic tools rarely support patient-history-aware predictions or longitudinal trend analysis across repeated assessments.

1.2 Project Objectives

To address these challenges, we have defined the following core objectives:

1. **Develop an Explainable Deep Learning Model:** Architect and train a hybrid CNN-LSTM-Attention model on the UCI Cleveland Heart Disease dataset. This model will incorporate multiple Explainable AI (XAI) methods, specifically Grad-CAM, Attention weights, and SHAP (Shapley Additive exPlanations), to decode the model's reasoning.
2. **Deliver an Interactive Clinical Dashboard:** Create a responsive, premium Single Page Application (SPA) using HTML5, CSS3, Bootstrap 5, and JavaScript. This dashboard will allow clinicians to input physiological parameters and visualize predictions via Plotly.js interactive charts.
3. **Automate Clinical Reporting:** Generate automated PDF reports featuring AI-driven narrative summaries, feature-by-feature clinical interpretations, and guideline-referenced evaluations.
4. **Implement Patient Management Architecture:** Design a robust backend using Flask (Python) and a MySQL database to securely track longitudinal patient records.

Chapter 2

Methodology and Design

2.1 System Architecture

Our proposed solution combines deep learning engineering with clinical interpretability, focusing on robust pipeline design and multi-method XAI.

2.1.1 High-Level Design and Modeling

During the initial phase of the project, significant effort was dedicated to defining actor modeling, system flow, and architectural blueprints.

- **Deep Learning Pipeline:** The core prediction engine is designed around a CNN-LSTM-Attention network. Clinical data will be transformed into spatiotemporal formats using Gramian Angular Summation Field (GASF) image encoding for robust feature extraction.
- **Multi-Method Explainability Engine:** The architecture maps parallel processing flows to compute SHAP values, Grad-CAM 2D heatmaps, and Attention weights simultaneously after the CNN-LSTM inference.
- **UML Modeling:** We designed a comprehensive UML Use Case Diagram establishing boundaries and main features for the primary actor (Clinician) and secondary actors (Patient, ML Engine, MySQL Database). Features include predicting risk, viewing XAI charts, performing Exploratory Data Analysis (EDA), and downloading PDF reports.
- **Activity Diagram:** A 4-swimlane UML Activity Diagram was produced to clarify responsibilities across the Frontend (Clinician), Middleware (Flask Web Server), Backend (ML Prediction Engine), and Storage (MySQL).

2.1.2 Technical Stack

The project leverages a modern, high-performance technology stack:

- **Frontend Engineering:** HTML5, modern CSS3 (with custom animations), Bootstrap 5, and Vanilla JavaScript.

- **Data Visualization & Aesthetics:** Plotly.js for rendering interactive charts, combined with Three.js and Vanta.js for dynamic, hardware-accelerated cybernetic backgrounds.
- **Audio Feedback:** Web Audio API for synthesizing Sci-Fi diagnostic audio cues without external audio files.
- **Machine Learning:** TensorFlow/Keras for the CNN architecture, scikit-learn for data preprocessing, and pyts for Gramian Angular Summation Field (GASF) transformations.
- **Backend & Integration (Future Work):** Flask (Python) for the RESTful API bridge between the ML engine and the Frontend SPA.

2.2 Implementation Details

2.2.1 Module 1: UI Prototyping and Frontend Architecture

The frontend Single Page Application (SPA) prototype has been successfully implemented, focusing heavily on a premium medical cybernetic aesthetic. The UI was designed to minimize cognitive load while maximizing clinical engagement.

Visual Aesthetics and Theming

We integrated glassmorphism techniques—utilizing semi-transparent backgrounds with backdrop blur filters—to provide tactile feedback on diagnostic panels. A sophisticated dark theme (#050811) was chosen to reduce eye strain for clinicians. The background features an interactive 3D Vanta.js network animation that dynamically shifts colors (from neon blue to danger red) based on the computed patient risk probability. Custom typography utilizes Google Fonts **Outfit** and **Plus Jakarta Sans** for high legibility in clinical settings. Furthermore, bespoke CSS animations such as CRT scanlines and floating 3D depth (`translateZ`) elements enhance the futuristic, premium feel of the dashboard.

Dynamic Input Interfaces

The application architecture relies on three primary SPA states manipulated dynamically via JavaScript:

- **Parameter Input Grid:** A responsive 3-column layout captures 13 critical physiological features (e.g., Age, Sex, Chest Pain Type, Resting BP, Cholesterol, Fasting Blood Sugar, Resting ECG, Max Heart Rate, Exercise Induced Angina, ST Depression, ST Slope, Number of Major Vessels, Thalassemia).
- **Dual-Input Controls & Clinical Ranges:** Features like Age, BP, and Cholesterol utilize synchronized range sliders and numerical inputs. Custom JavaScript logic dynamically updates "Glowing Badges" and clinical reference texts (e.g., tagging BP ≥ 140 as "Hypertension" or Cholesterol ≥ 200 as "Desirable") in real-time as the user adjusts the inputs.

- **Patient Management Mockups:** We implemented UI components for toggling between "New" and "Existing" patients, including an autocomplete datalist structure for searching patient hospital IDs.

Simulated Diagnostic Pipeline

To simulate the backend processing delay, we engineered a highly immersive "System Processing" state:

- **Visual Feedback:** A non-blocking diagnostic scan overlay features complex CSS radar pulse (`radar-pulse`) and sweep animations alongside an SVG sinus rhythm drawing animation.
- **Scrolling Terminal:** A cascading scrolling log terminal outputs simulated computation stages (e.g., "[SYSTEM] Ingesting Telemetry Data...", "[AI] Converting 1D telemetry array to 2D Gramian Field (GASF)...").
- **Audio Synthesis:** We implemented a custom `SciFiAudioController` using the HTML5 Web Audio API. This synthesizes continuous sonar sweeps, data processing ticks, and a success chime upon completion, entirely through code.

Results Dashboard and XAI Visualization

The final SPA state is a comprehensive analytics module designed to present the AI's findings:

- **Risk Probability:** Displays a dynamic progress bar and a CSS-animated beating heart whose pulse rate correlates with the predicted risk percentage.
- **Performance Metrics:** UI containers for displaying Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- **Plotly.js Integrations:** The dashboard includes initialized Plotly.js containers prepared to render the "XAI Consensus Profile" (comparing Global Attention, Local Grad-CAM, and Patient SHAP weights) and a detailed "2D Grad-CAM Heatmap".
- **AI Narrative Summary:** A dedicated section for generating human-readable clinical narratives based on the model's primary diagnostic drivers.

Client-side HTML5 validations and a robust mock JSON data generator have been configured to ensure the frontend prototype is fully operational and demonstrable prior to backend integration.

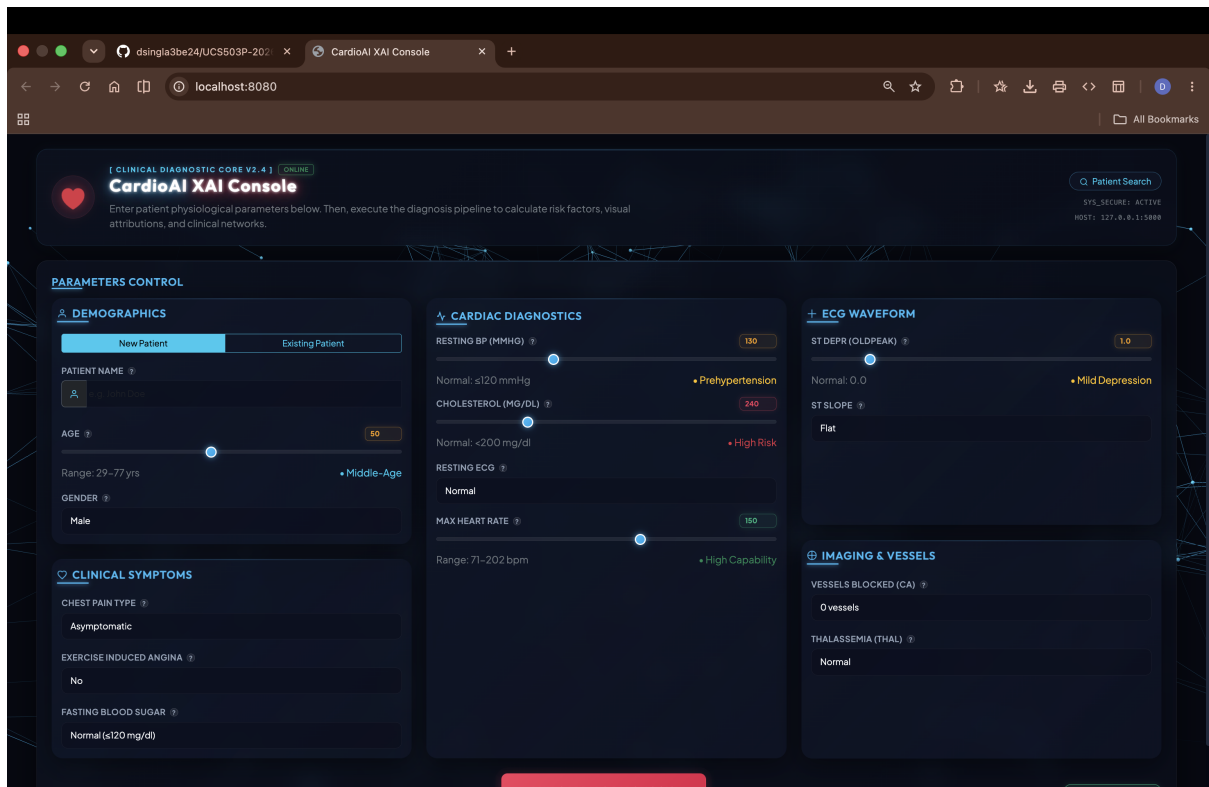


Figure 2.1: Initial Data Input Dashboard with 3D Cybernetic Background.

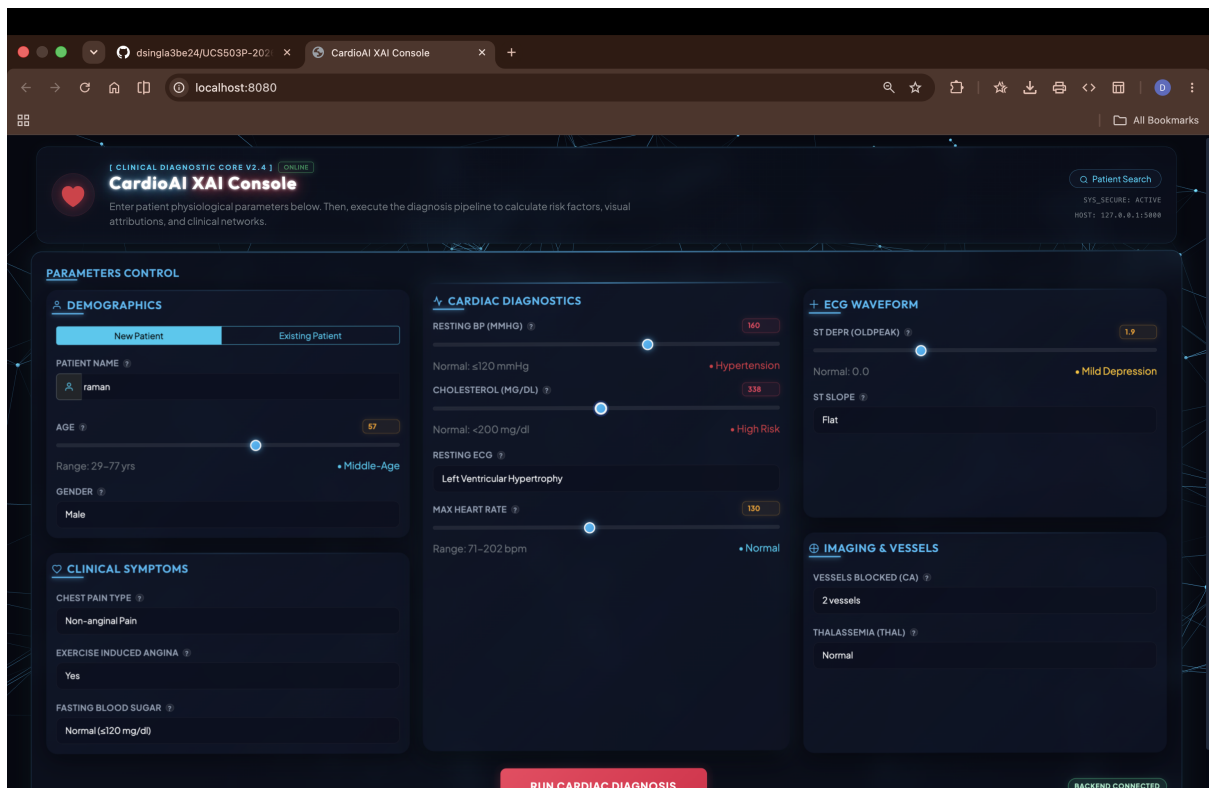


Figure 2.2: Active Parameter Entry utilizing synchronized sliders and glowing clinical badges.

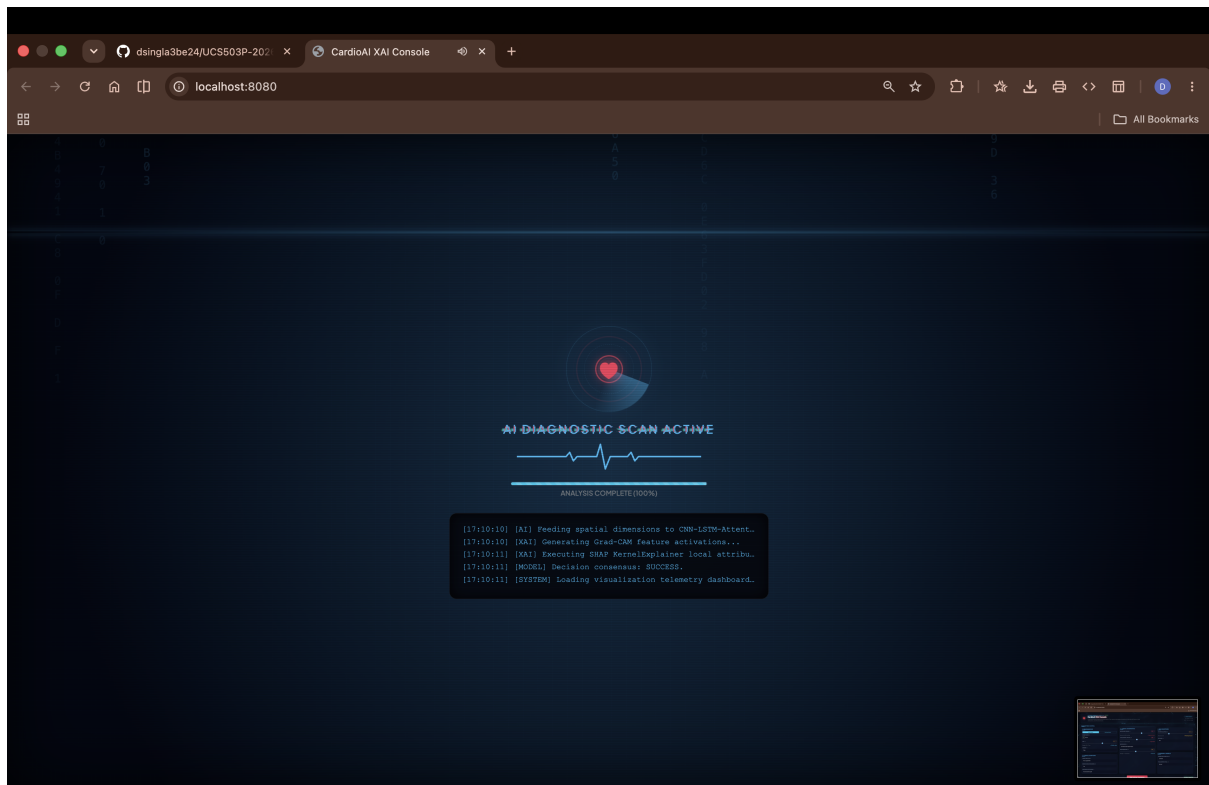


Figure 2.3: System Processing State: Radar sweep and diagnostic overlay.

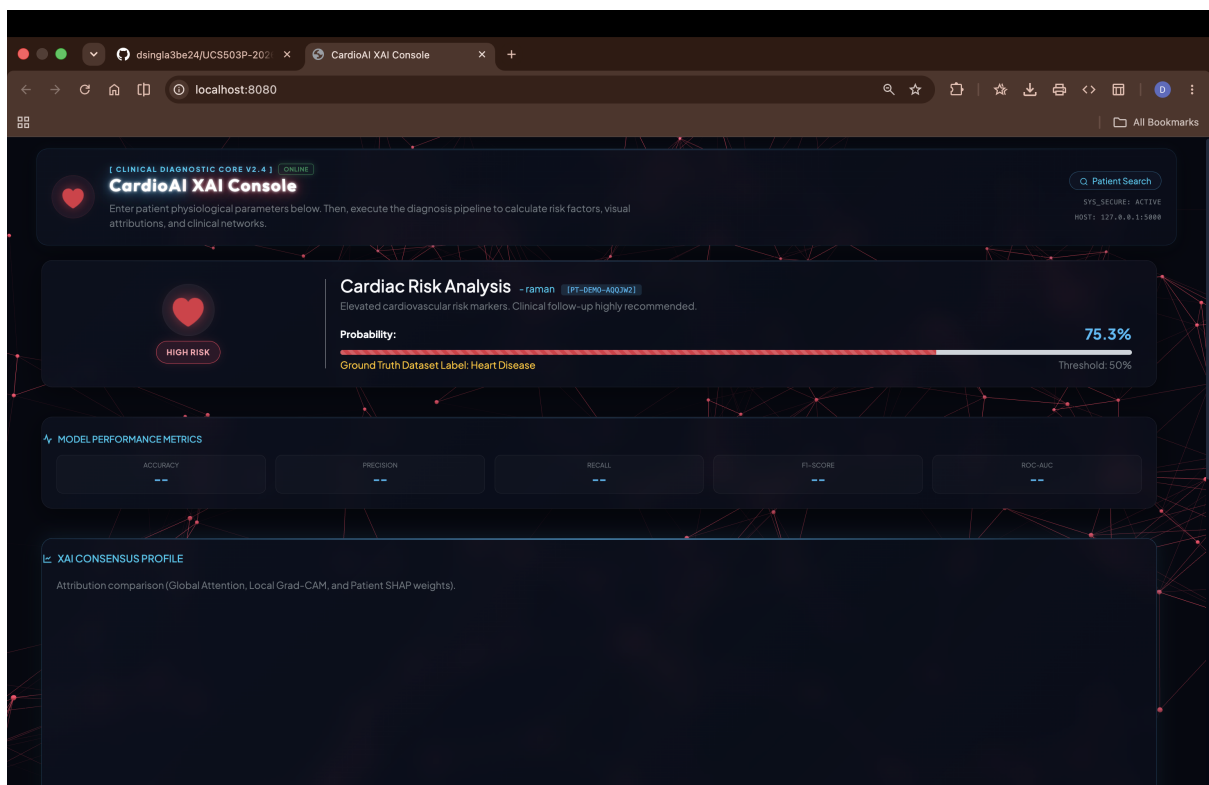


Figure 2.4: Results Dashboard: Predicted Risk Probability and Patient Metrics.

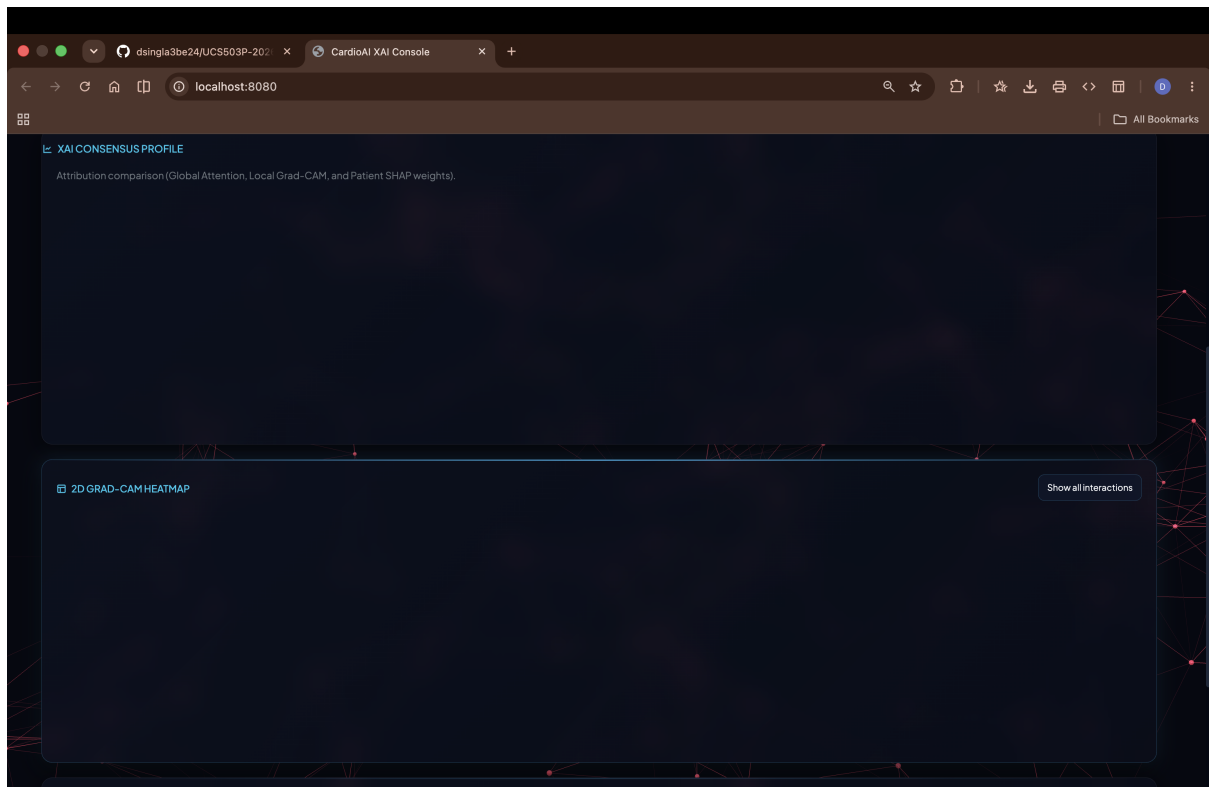


Figure 2.5: XAI Consensus Profile: Comparing Grad-CAM, SHAP, and Attention Heatmaps.

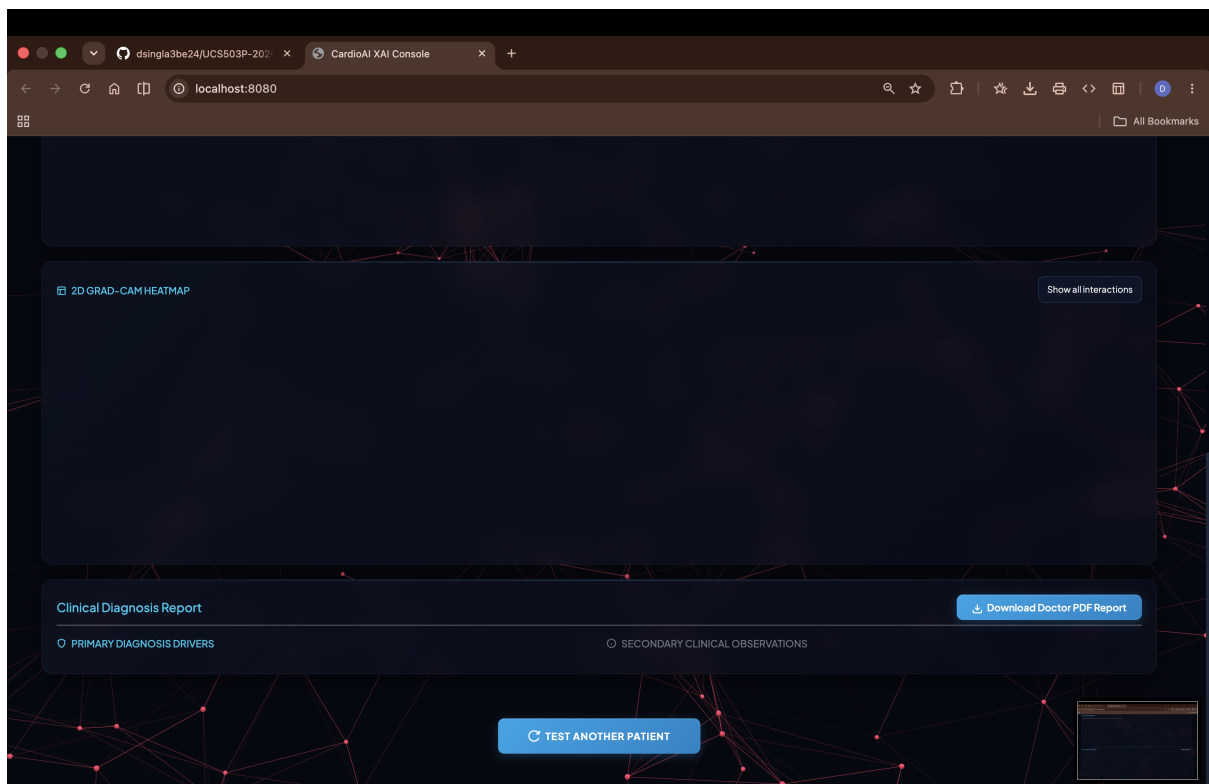


Figure 2.6: Results Dashboard: Patient Narrative Summary and PDF Report Generation.

2.2.2 Module 2: Deep Learning Baseline Pipeline (CNN)

To establish the core prediction engine, we implemented the foundational machine learning backend up to the Convolutional Neural Network (CNN) phase:

- **Data Ingestion & Scaling:** The UCI Heart Disease dataset was ingested and all 13 clinical features were standardized using MinMaxScaler to ensure uniform gradient propagation.
- **Mathematical Transformation (GASF):** Since CNNs are optimized for 2D spatial data, we transformed the 1D tabular arrays into 2D Gramian Angular Summation Field (GASF) images. This cutting-edge technique encodes temporal/feature correlations into a visual format.
- **CNN Architecture:** A Keras Sequential model was constructed using Conv2D and MaxPooling2D layers to extract spatial features from the GASF matrix, culminating in a Dense output layer with a Sigmoid activation for binary risk classification.
- **Initial Evaluation:** The baseline CNN was trained successfully, with loss curves and confusion matrices plotted to verify model convergence before the future integration of LSTM and Explainable AI (Grad-CAM/SHAP) components.

2.3 System Modeling and Project Diagrams

To provide a comprehensive blueprint of the Cardio AI system, several architectural and behavioral models were developed.

2.3.1 Project Scheduling and Timeline (Gantt Chart)

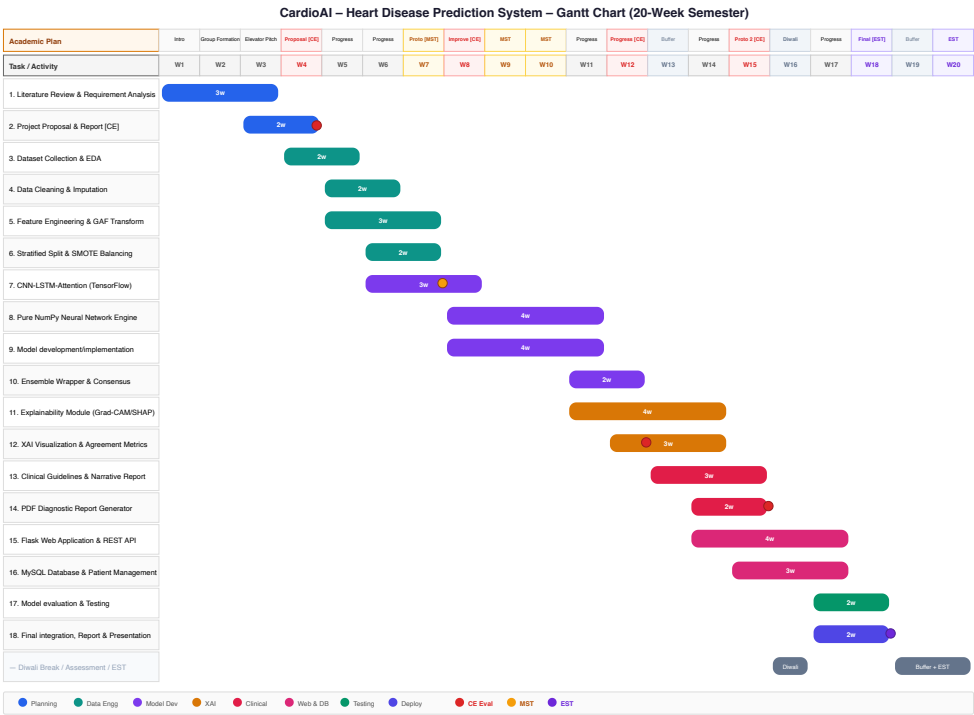


Figure 2.7: Project Scheduling and Timeline (Gantt Chart)

2.3.2 UML Use Case Diagram

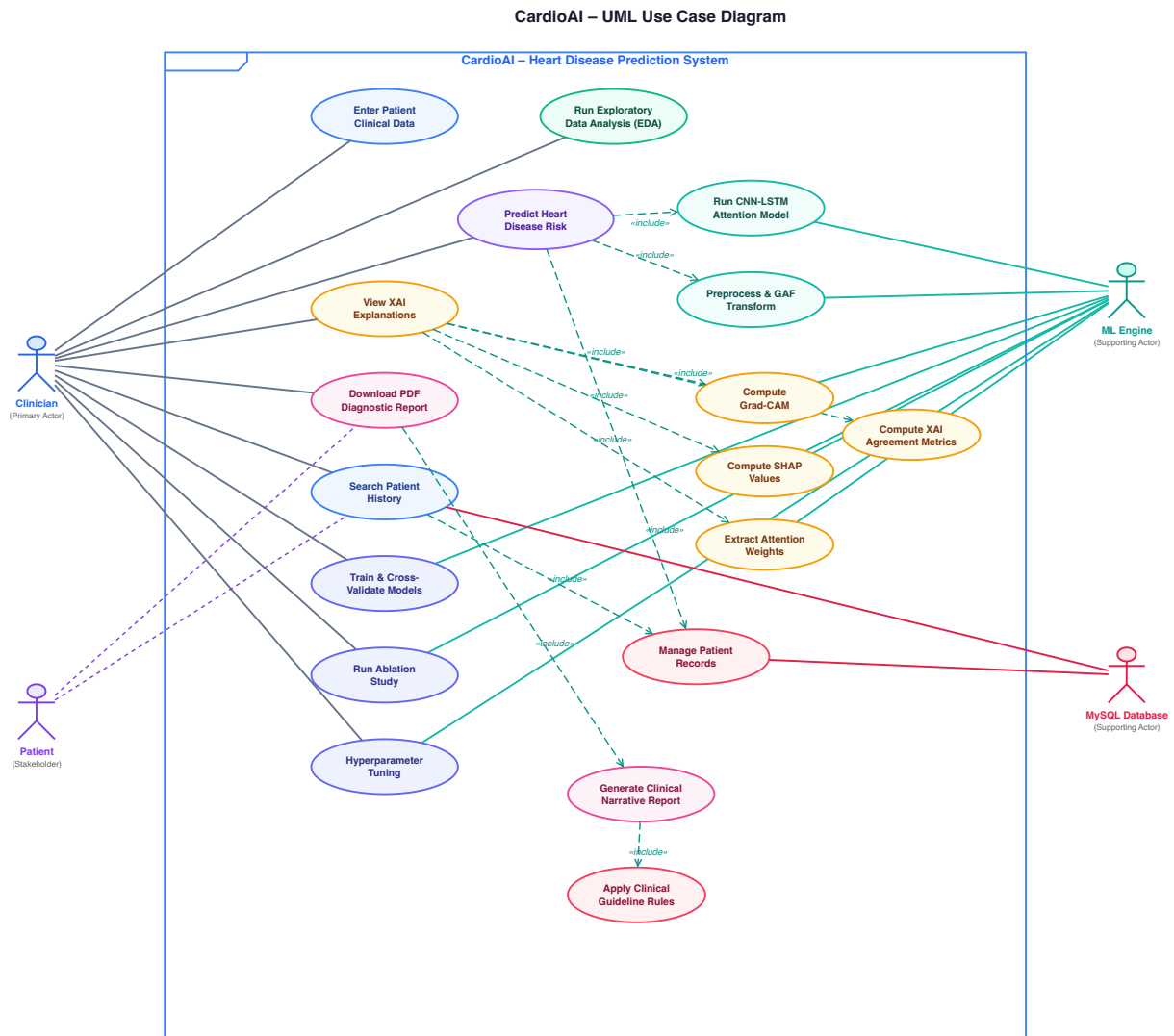


Figure 2.8: UML Use Case Diagram identifying system boundaries and primary actors.

2.3.3 Use Case Specifications

The following sections detail the specific use case templates that define the interactions between actors and the Cardio AI system.

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Enter Patient Clinical Data
2. Abbreviated Title	Enter Clinical Data
3. Use Case Id	UC-01
4. Actors	Clinician (Primary)
5. Description	The Clinician enters 13 physiological features for a patient (Age, Gender, Chest Pain Type, Resting BP, Cholesterol, Fasting Blood Sugar, Resting ECG, Max Heart Rate, Exercise Angina, ST Depression, ST Slope, Number of Major Vessels, Thalassemia) into the CardioAI web dashboard input form.
5.1. Pre Conditions	1. Clinician has access to the CardioAI web dashboard. 2. Patient's clinical data is available from medical records.
5.2. Task Sequence	1. Clinician opens the CardioAI dashboard. 2. System displays the Parameter Input form with 3 columns. 3. Clinician enters patient data using input fields, sliders, and dropdowns. 4. System validates inputs in real-time (range checks, required fields). 5. System confirms all 13 features are populated and valid.
5.3. Post Conditions	1. Patient data is captured in the frontend form. 2. Data is ready for prediction submission or record storage.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.9: Use Case: Enter Clinical Data

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Run Exploratory Data Analysis (EDA)
2. Abbreviated Title	Run EDA
3. Use Case Id	UC-02
4. Actors	Clinician (Primary), ML Engine (Supporting)
5. Description	The Clinician initiates an EDA pipeline on the Cleveland Heart Disease dataset to visualize distributions, correlations, missing values, and feature importance before training the model.
5.1. Pre Conditions	1. The Cleveland dataset is loaded and accessible. 2. EDA dependencies (Pandas, Matplotlib, Seaborn) are installed.
5.2. Task Sequence	1. Clinician triggers the EDA module via CLI or dashboard. 2. System loads and profiles the dataset. 3. System generates statistical summaries (mean, std, missing %). 4. System renders distribution plots for each feature. 5. System renders a correlation heatmap. 6. System highlights key insights (class imbalance, outliers).
5.3. Post Conditions	1. Visualizations (histograms, heatmaps, box plots) are generated. 2. Key statistical insights are available for model training decisions.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.10: Use Case: Run EDA

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Predict Heart Disease Risk
2. Abbreviated Title	Predict Risk
3. Use Case Id	UC-03
4. Actors	Clinician (Primary), ML Engine (Supporting), MySQL Database (Supporting)
5. Description	The Clinician submits validated patient data to obtain a binary heart disease risk prediction (HIGH/LOW) with a probability score from the CNN-LSTM-Attention deep learning model. Includes preprocessing, model inference, and record storage.
5.1. Pre Conditions	1. All 13 patient features are entered and validated (UC-01). 2. A trained model checkpoint exists and is loaded. 3. Backend Flask server is running.
5.2. Task Sequence	1. Clinician clicks 'Run Diagnostic Scan' after entering patient data. 2. System transitions to the Processing Overlay (scanning animation). 3. System invokes UC-10: Preprocess and GAF Transform on the input data. 4. System invokes UC-11: Run CNN-LSTM Attention Model for inference. 5. System invokes UC-15: Manage Patient Records to store the result. 6. System transitions to the Results Dashboard. 7. System displays risk label (HIGH/LOW), probability score, and metrics.
5.3. Post Conditions	1. Risk prediction (probability and label) is displayed on the dashboard. 2. Prediction record is saved to the MySQL database. 3. XAI analysis is triggered automatically.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.11: Use Case: Predict Risk

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	View XAI Explanations
2. Abbreviated Title	View XAI
3. Use Case Id	UC-04
4. Actors	Clinician (Primary), ML Engine (Supporting)
5. Description	After a prediction, the Clinician views Explainable AI visualizations that explain why the model made a specific prediction, using three independent XAI methods (Grad-CAM, SHAP, Attention Weights) and their inter-method agreement metrics.
5.1. Pre Conditions	1. A prediction has been successfully completed (UC-03). 2. XAI computation modules (Grad-CAM, SHAP, Attention) are available.
5.2. Task Sequence	1. System invokes UC-12: Compute Grad-CAM to generate the heatmap. 2. System invokes UC-13: Compute SHAP Values for feature importance. 3. System invokes UC-14: Extract Attention Weights for attention distribution. 4. System invokes UC-18: Compute XAI Agreement Metrics. 5. System renders all visualizations in Plotly.js chart containers. 6. Clinician interacts with charts (hover, zoom, filter features).
5.3. Post Conditions	1. Grad-CAM heatmap, SHAP bar chart, Attention visualization are rendered. 2. XAI agreement score is displayed. 3. Clinician understands why the model made the prediction.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.12: Use Case: View XAI

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Download PDF Diagnostic Report
2. Abbreviated Title	Download PDF Report
3. Use Case Id	UC-05
4. Actors	Clinician (Primary), Patient (Stakeholder)
5. Description	The Clinician generates and downloads a comprehensive PDF report containing patient demographics, prediction results, XAI visualizations, and a clinical narrative summary for the patient's medical record.
5.1. Pre Conditions	1. A prediction (UC-03) and XAI analysis (UC-04) have been completed. 2. PDF generation library is available.
5.2. Task Sequence	1. Clinician clicks 'Download Report' on the Results Dashboard. 2. System invokes UC-16: Generate Clinical Narrative Report. 3. System compiles patient data, prediction, XAI charts, and narrative. 4. System generates the formatted PDF document. 5. Browser triggers a file download to the Clinician's device.
5.3. Post Conditions	1. A formatted PDF diagnostic report is downloaded. 2. Report can be shared with the Patient or added to medical records.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.13: Use Case: Download PDF Report

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Search Patient History
2. Abbreviated Title	Search History
3. Use Case Id	UC-06
4. Actors	Clinician (Primary), Patient (Stakeholder), MySQL Database (Supporting)
5. Description	The Clinician searches for a patient's historical records including past predictions, risk trends over time, and previously generated diagnostic reports stored in the MySQL database.
5.1. Pre Conditions	1. Patient records exist in the MySQL database. 2. Database connection is active.
5.2. Task Sequence	1. Clinician navigates to the Patient History section. 2. Clinician enters patient identifier (Name or ID). 3. System invokes UC-15: Manage Patient Records to query the database. 4. System retrieves and displays matching records. 5. System renders a timeline or trend chart of past risk predictions.
5.3. Post Conditions	1. Patient's longitudinal history is displayed. 2. Past predictions and trend data are available for review.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.14: Use Case: Search History

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Train and Cross-Validate Models
2. Abbreviated Title	Train Models
3. Use Case Id	UC-07
4. Actors	Clinician (Primary), ML Engine (Supporting)
5. Description	The Clinician initiates a model training pipeline with k-fold cross-validation on the heart disease dataset to produce a deployable CNN-LSTM-Attention model with validated performance metrics.
5.1. Pre Conditions	1. Preprocessed dataset is available. 2. Training hyperparameters are configured. 3. GPU/CPU resources are available.
5.2. Task Sequence	1. Clinician configures training parameters (epochs, learning rate, k-folds). 2. Clinician initiates training via CLI or dashboard. 3. ML Engine loads and preprocesses the dataset. 4. ML Engine performs k-fold cross-validation. 5. ML Engine trains the CNN-LSTM-Attention model on each fold. 6. System logs per-fold and average metrics (Accuracy, F1, ROC-AUC). 7. System saves the best model checkpoint.
5.3. Post Conditions	1. A trained model checkpoint is saved to disk. 2. Cross-validation metrics are logged and available for review.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.15: Use Case: Train Models

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Run Ablation Study
2. Abbreviated Title	Ablation Study
3. Use Case Id	UC-08
4. Actors	Clinician (Primary), ML Engine (Supporting)
5. Description	The Clinician runs ablation experiments to evaluate the contribution of individual model components (CNN branch, LSTM branch, Attention mechanism) to the overall prediction performance.
5.1. Pre Conditions	1. A fully trained CNN-LSTM-Attention model exists. 2. Test dataset is available for evaluation.
5.2. Task Sequence	1. Clinician selects components to ablate (e.g., remove Attention layer). 2. System creates a modified model variant with the component disabled. 3. ML Engine evaluates the modified model on the test set. 4. System records performance metrics for the ablated variant. 5. Steps 2-4 repeat for each ablation configuration. 6. System generates a comparison table/chart of all variants.
5.3. Post Conditions	1. Comparative metrics (with/without each component) are generated. 2. Component contribution to overall accuracy is quantified.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.16: Use Case: Ablation Study

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Hyperparameter Tuning
2. Abbreviated Title	Hyperparameter Tuning
3. Use Case Id	UC-09
4. Actors	Clinician (Primary), ML Engine (Supporting)
5. Description	The Clinician configures and runs automated hyperparameter search (grid search or Bayesian optimization) to find optimal model configurations for learning rate, dropout, hidden units, batch size, etc.
5.1. Pre Conditions	1. Training pipeline is functional (UC-07). 2. Search space is defined for target hyperparameters.
5.2. Task Sequence	1. Clinician defines the hyperparameter search space. 2. Clinician selects search strategy (Grid / Random / Bayesian). 3. ML Engine iterates through hyperparameter combinations. 4. Each combination is evaluated via cross-validation. 5. System ranks results by validation ROC-AUC. 6. System outputs the best configuration and saves the model.
5.3. Post Conditions	1. Optimal hyperparameters are identified and logged. 2. Best model corresponding to optimal config is saved.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.17: Use Case: Hyperparameter Tuning

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Preprocess and GAF Transform
2. Abbreviated Title	Preprocess GAF
3. Use Case Id	UC-10
4. Actors	ML Engine (Supporting)
5. Description	The ML Engine preprocesses raw patient data (normalization, encoding, missing value handling) and applies the Gramian Angular Field (GAF) transformation to convert tabular features into 2D image representations for the CNN branch.
5.1. Pre Conditions	1. Raw patient feature vector (13 features) is received from UC-01. 2. Preprocessing pipeline configuration is loaded.
5.2. Task Sequence	1. System receives the raw 13-feature vector. 2. System applies min-max normalization scaled to [-1, 1]. 3. System encodes categorical features (Gender, Chest Pain Type, etc.). 4. System applies Gramian Angular Field transformation. 5. System outputs the normalized vector and GAF image matrix.
5.3. Post Conditions	1. Normalized feature vector is ready for the LSTM branch. 2. GAF image matrix is ready for the CNN branch.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.18: Use Case: Preprocess GAF

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Run CNN-LSTM Attention Model
2. Abbreviated Title	Run Model Inference
3. Use Case Id	UC-11
4. Actors	ML Engine (Supporting)
5. Description	The ML Engine runs forward inference through the hybrid CNN-LSTM-Attention architecture using preprocessed inputs (normalized vector + GAF image) to produce a binary heart disease risk probability score.
5.1. Pre Conditions	1. Preprocessed data is available from UC-10. 2. Trained model checkpoint is loaded into memory.
5.2. Task Sequence	1. CNN branch processes the GAF image to extract spatial features. 2. LSTM branch processes the feature sequence for sequential patterns. 3. Attention mechanism computes weighted feature importance. 4. Fused representation passes through the classification head. 5. System outputs sigmoid probability and risk label (HIGH/LOW).
5.3. Post Conditions	1. Binary classification output (probability + label) is produced. 2. Internal activations are stored for XAI computation.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.19: Use Case: Run Model Inference

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Compute Grad-CAM
2. Abbreviated Title	Compute Grad-CAM
3. Use Case Id	UC-12
4. Actors	ML Engine (Supporting)
5. Description	The ML Engine computes Gradient-weighted Class Activation Mapping (Grad-CAM) on the CNN branch's final convolutional layer to produce a heatmap highlighting which regions of the GAF image most influenced the prediction.
5.1. Pre Conditions	1. Model inference (UC-11) has been completed. 2. Gradients are available (model in eval mode with gradient hooks).
5.2. Task Sequence	1. System registers gradient hooks on the target convolutional layer. 2. System performs a forward pass with the patient input. 3. System computes gradients of predicted class w.r.t. target activations. 4. System performs global average pooling to get channel weights. 5. System computes weighted sum of activation maps and applies ReLU. 6. System maps heatmap regions back to clinical feature names.
5.3. Post Conditions	1. A 2D Grad-CAM heatmap overlay is generated. 2. Heatmap is mapped to feature names for clinical interpretation.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.20: Use Case: Compute Grad-CAM

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Compute SHAP Values
2. Abbreviated Title	Compute SHAP
3. Use Case Id	UC-13
4. Actors	ML Engine (Supporting)
5. Description	The ML Engine computes SHAP (SHapley Additive exPlanations) values to quantify each of the 13 clinical features' marginal contribution to the model's prediction output.
5.1. Pre Conditions	1. Model inference (UC-11) has been completed. 2. Background dataset (training sample) is available for SHAP explainer.
5.2. Task Sequence	1. System initializes DeepSHAP explainer with background data. 2. System computes SHAP values for the current patient input. 3. System ranks features by absolute SHAP value. 4. System generates positive/negative contribution indicators. 5. System outputs sorted feature importance list for visualization.
5.3. Post Conditions	1. A vector of 13 SHAP values (one per feature) is produced. 2. Feature importance ranking is ready for bar/waterfall chart rendering.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.21: Use Case: Compute SHAP

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Extract Attention Weights
2. Abbreviated Title	Extract Attention
3. Use Case Id	UC-14
4. Actors	ML Engine (Supporting)
5. Description	The ML Engine extracts the learned attention weight distribution from the Attention layer, revealing which clinical features the model focused on most during the heart disease prediction.
5.1. Pre Conditions	1. Model inference (UC-11) has been completed. 2. Attention layer weights are stored during the forward pass.
5.2. Task Sequence	1. System retrieves raw attention scores from the model's attention layer. 2. System applies softmax normalization. 3. System maps each weight to the corresponding clinical feature name. 4. System outputs the attention distribution for visualization.
5.3. Post Conditions	1. A normalized attention weight vector (sums to 1.0) is produced. 2. Weights are mapped to feature names for clinical display.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.22: Use Case: Extract Attention

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Manage Patient Records
2. Abbreviated Title	Manage Records
3. Use Case Id	UC-15
4. Actors	MySQL Database (Supporting)
5. Description	The system performs CRUD (Create, Read, Update, Delete) operations on patient records in the MySQL database, including storing prediction results, updating demographics, and retrieving longitudinal history.
5.1. Pre Conditions	1. MySQL database server is running and accessible. 2. Database schema (patients, predictions tables) is initialized.
5.2. Task Sequence	1. Create: System inserts a new patient record with demographics and prediction. 2. Read: System retrieves a patient record by ID or search criteria. 3. Update: System appends new prediction results to existing patient history. 4. Delete: System removes a patient record (soft delete with audit trail).
5.3. Post Conditions	1. Patient record is created, updated, retrieved, or deleted as requested. 2. Database maintains referential integrity and audit trail.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.23: Use Case: Manage Records

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Generate Clinical Narrative Report
2. Abbreviated Title	Generate Narrative
3. Use Case Id	UC-16
4. Actors	ML Engine (Supporting)
5. Description	The system generates a human-readable clinical narrative summarizing the prediction results, key risk factors identified by XAI methods, and relevant clinical context in natural language for inclusion in the PDF report.
5.1. Pre Conditions	1. Prediction (UC-03) and XAI analysis (UC-04) are complete. 2. Narrative template is available.
5.2. Task Sequence	1. System collects patient demographics, prediction result, and top-3 features. 2. System invokes UC-17: Apply Clinical Guideline Rules. 3. System generates narrative using template-based text generation. 4. System outputs the formatted clinical summary text.
5.3. Post Conditions	1. A structured clinical narrative text is generated. 2. Narrative includes guideline-based context and risk factor explanation.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.24: Use Case: Generate Narrative

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Apply Clinical Guideline Rules
2. Abbreviated Title	Apply Guidelines
3. Use Case Id	UC-17
4. Actors	ML Engine (Supporting)
5. Description	The system applies established clinical guideline rules (e.g., ACC/AHA cardiovascular risk thresholds) to provide evidence-based context alongside the ML prediction, enhancing clinician trust and interpretability.
5.1. Pre Conditions	1. Patient feature values are available. 2. Clinical guideline rule set is loaded in the system.
5.2. Task Sequence	1. System evaluates patient data against predefined clinical rules. 2. System flags values exceeding clinical thresholds (e.g., BP > 140). 3. System generates guideline-based recommendations. 4. System appends flags and recommendations to the narrative report.
5.3. Post Conditions	1. Clinical risk flags and guideline-based recommendations are generated. 2. Annotations are ready for inclusion in the diagnostic report.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.25: Use Case: Apply Guidelines

Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Compute XAI Agreement Metrics
2. Abbreviated Title	XAI Agreement
3. Use Case Id	UC-18
4. Actors	ML Engine (Supporting)
5. Description	The system computes agreement and consistency metrics across the three XAI methods (Grad-CAM, SHAP, Attention) to quantify how much they agree on which clinical features are most important, boosting explanation reliability.
5.1. Pre Conditions	1. All three XAI outputs (UC-12, UC-13, UC-14) are available. 2. Feature importance rankings from each method are normalized.
5.2. Task Sequence	1. System normalizes the feature importance rankings from each XAI method. 2. System computes pairwise rank correlation (Spearman's rho). 3. System computes top-k feature overlap across all 3 methods. 4. System computes an overall agreement score. 5. System outputs the score and a visual agreement matrix.
5.3. Post Conditions	1. An agreement score is computed and displayed on the dashboard. 2. Clinician can assess the reliability of XAI explanations.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)

Figure 2.26: Use Case: XAI Agreement

2.3.4 UML Activity Diagram

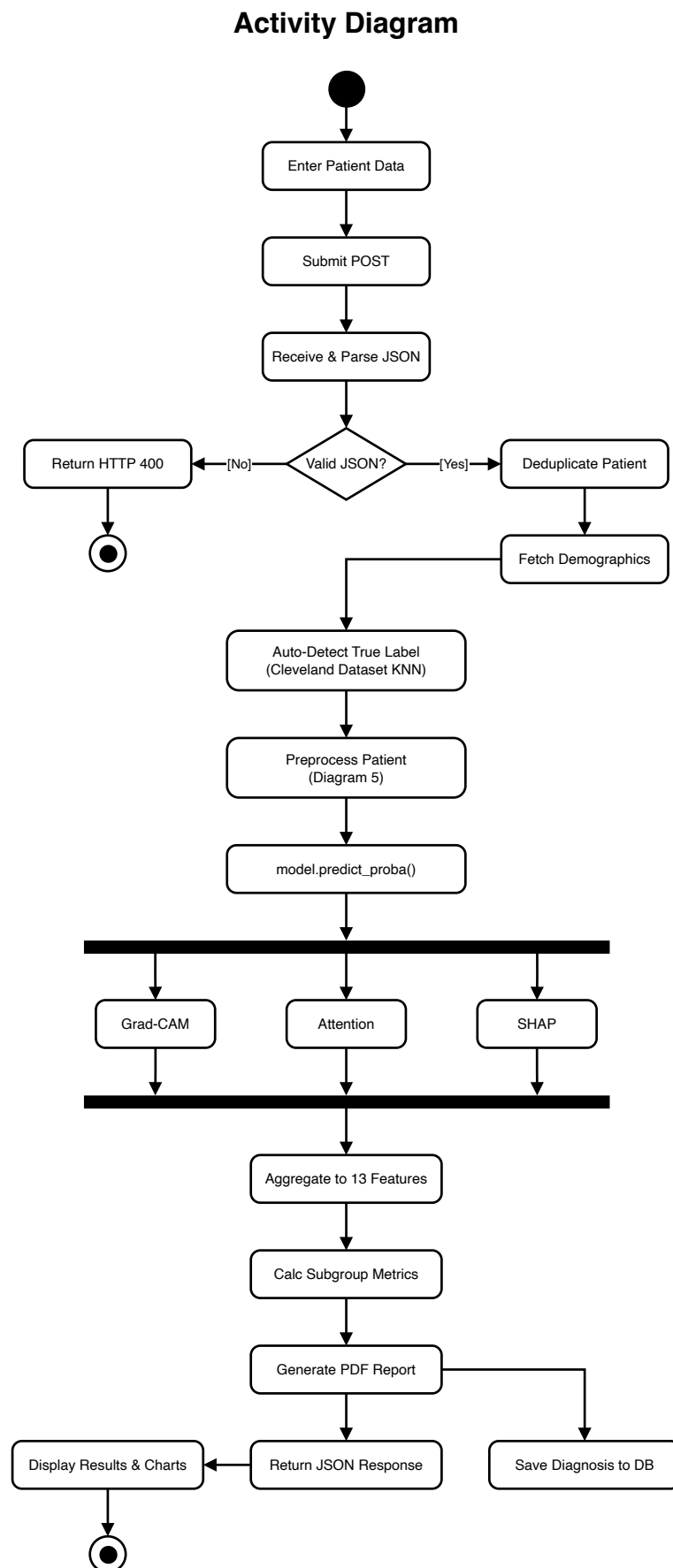


Figure 2.27: UML Activity Diagram mapping the procedural flow of the system.

2.3.5 UML Activity Diagram (Swimlanes)

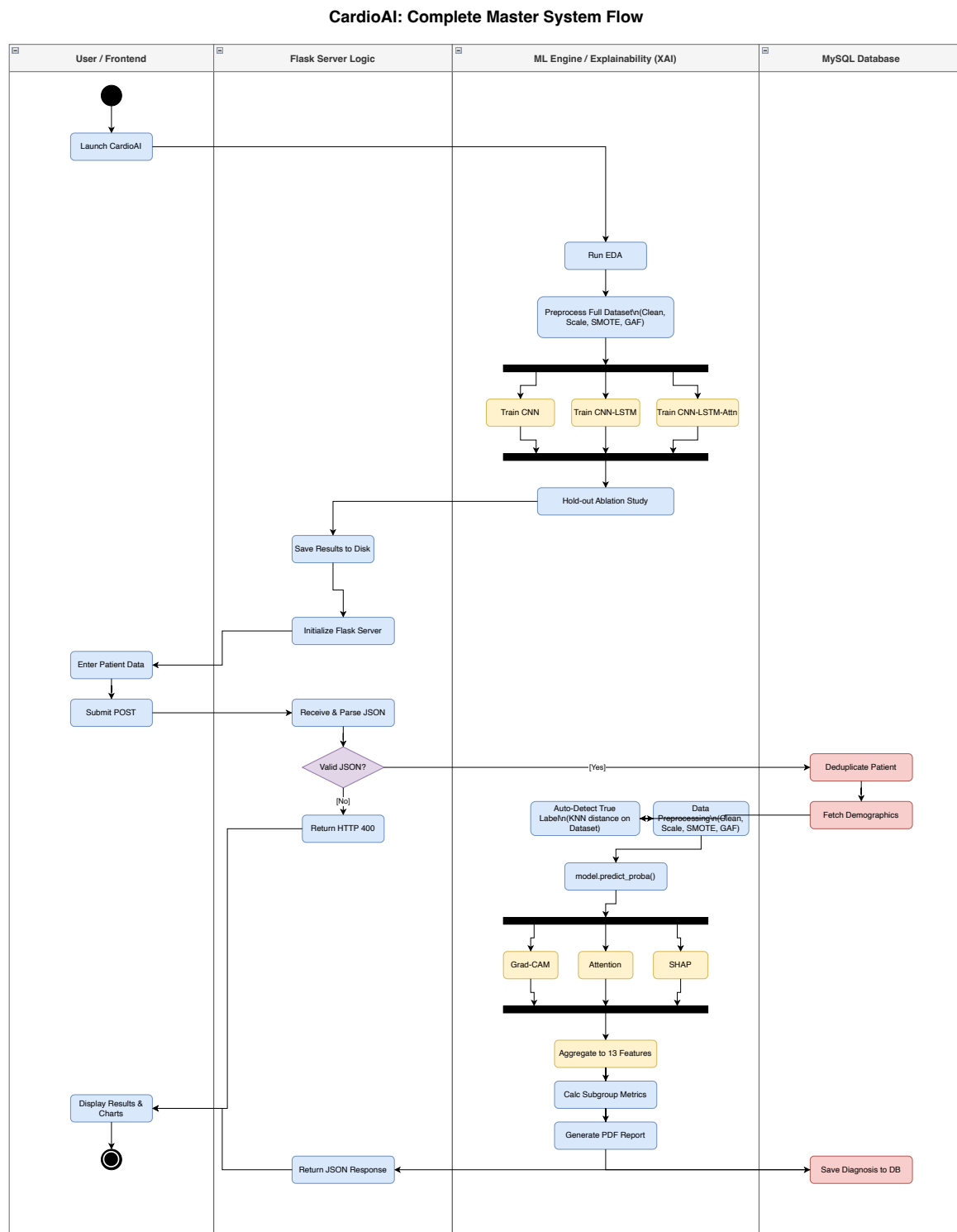


Figure 2.28: UML Activity Diagram mapped across specific actor swimlanes (Frontend, Middleware, Backend, DB).

2.3.6 UML Sequence Diagram

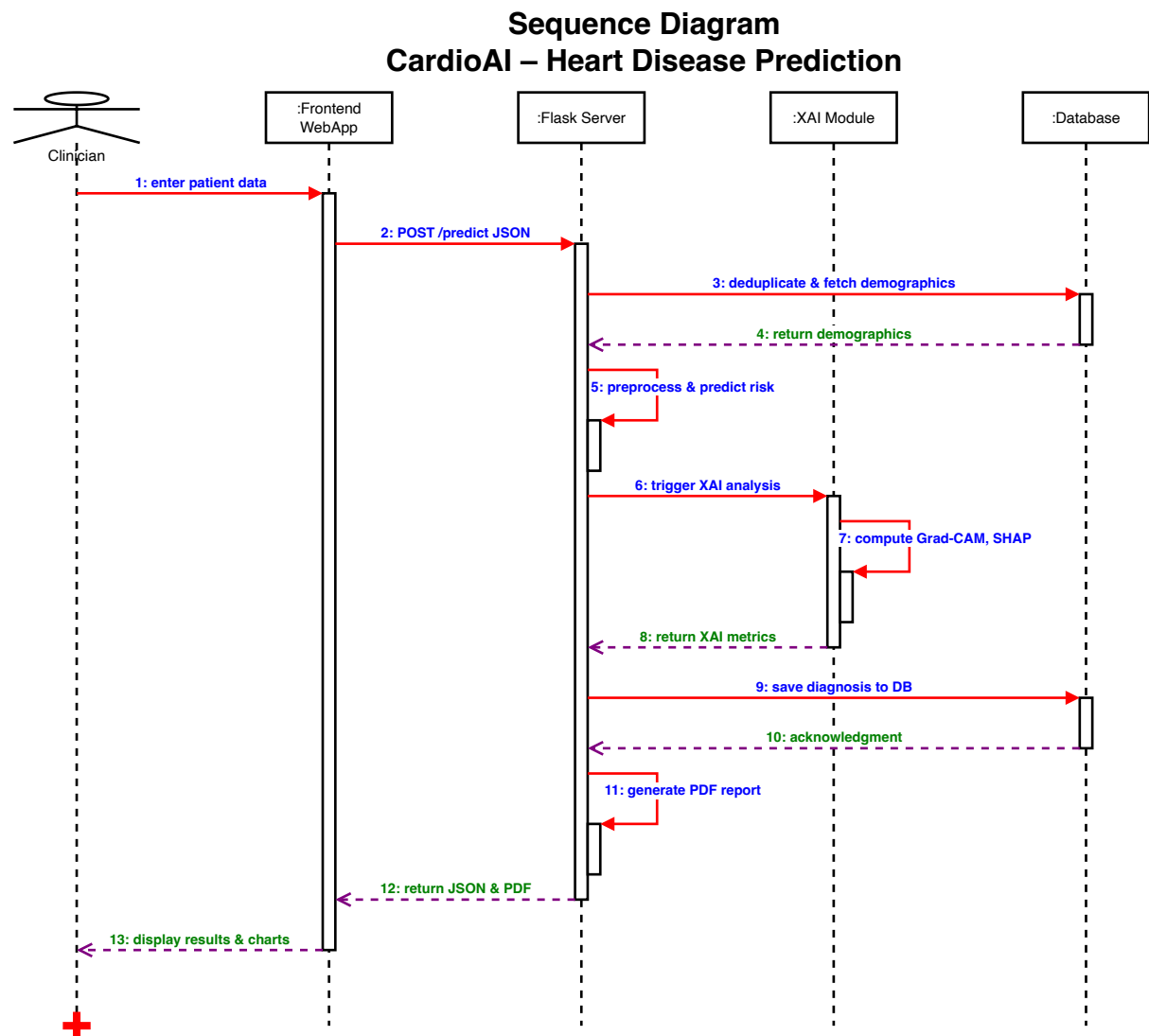


Figure 2.29: UML Sequence Diagram depicting the chronological interaction between system components.

2.3.7 UML Collaboration Diagram

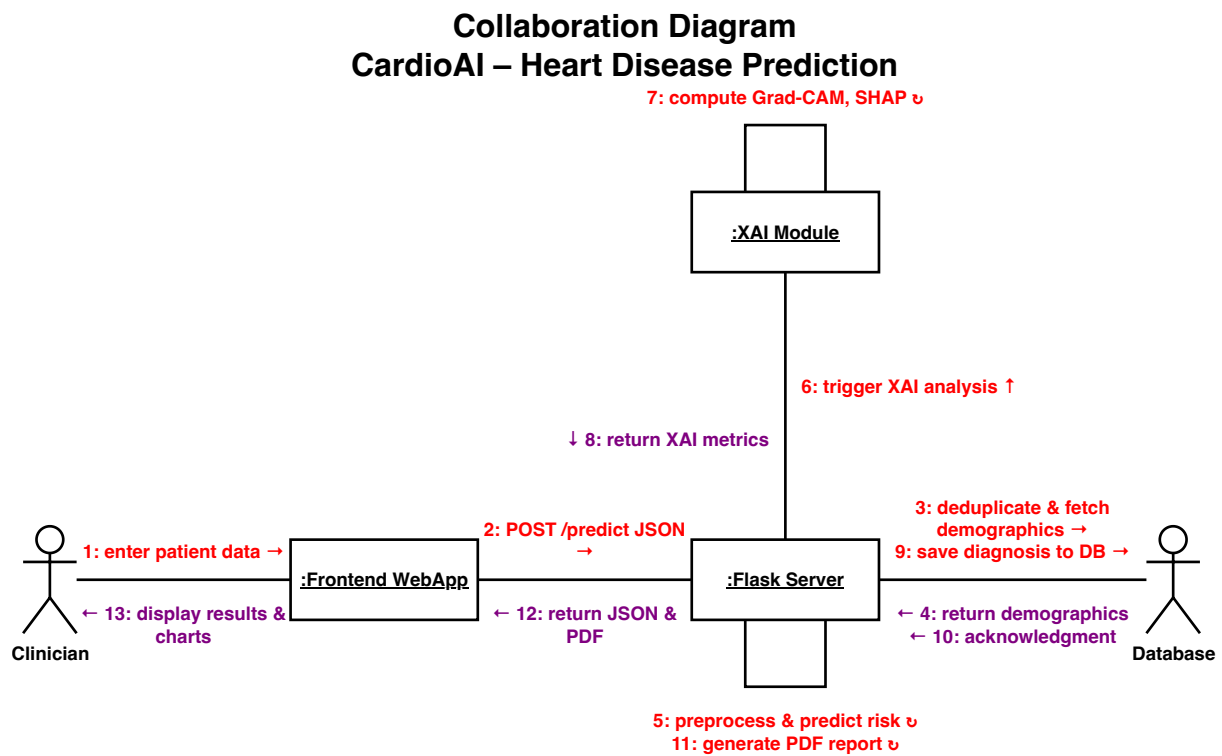


Figure 2.30: UML Collaboration Diagram highlighting the structural communication links between components.

Chapter 3

Progress Evaluation

3.1 Current Status and Validation

As this is the mid-semester evaluation stage, our efforts have successfully culminated in the completion of the foundational architecture, system modeling, a highly polished frontend prototype, and the core CNN machine learning baseline.

- **UML Design Validation:** The comprehensive UML models (Use Case, Activity, Sequence, and Collaboration) have been peer-reviewed and successfully validated as the logical blueprint for our distributed system.
- **UI/UX Testing:** The frontend components have undergone rigorous interface validation. The layout maintains responsiveness across desktop and tablet viewports, and input integrity is enforced via HTML5 attributes and JavaScript event listeners.
- **Mock Integration & Flow:** The SPA transitions seamlessly between data input, processing overlays (with synchronized audio-visual feedback), and the results dashboard. The JavaScript-based mock logic successfully populates the Plotly.js charts and UI badges, proving that the frontend is fully capable of rendering the data structures we intend to send from the backend.
- **CNN Baseline Validation:** The foundational machine learning pipeline was successfully structured and validated in a local Python environment, proving that the clinical dataset can be correctly scaled and transformed into GASF images before being processed by the Keras CNN layers.

3.2 Analysis of Progress

Our progress strictly adheres to our initial Gantt chart objectives for the Requirements, UI, and Baseline ML phases. The robust, state-driven design of the frontend SPA ensures that integrating the Flask REST API will require minimal to no UI refactoring.

Moving forward, the primary engineering challenge will shift to expanding the CNN baseline by adding the LSTM and Attention mechanisms to achieve our target AUC of ≥ 0.85 , and establishing the Flask bridge.

Chapter 4

Conclusion

4.1 Summary of Work

We have established a formidable core foundation for the Cardio AI prediction system. To date, we have:

- Defined the problem scope, focusing on clinical interpretability.
- Architected the proposed ML pipeline and finalized the technical stack.
- Completed comprehensive UML modeling (Use Case, Activity, Sequence, and Collaboration diagrams).
- Developed a detailed project schedule using a Gantt chart.
- Constructed a fully interactive, glassmorphic frontend SPA prototype, complete with dynamic clinical range validations, hardware-accelerated 3D backgrounds, and synthesized audio feedback.
- Engineered the foundational machine learning backend up to the Convolutional Neural Network (CNN) phase, successfully integrating clinical dataset ingestion, MinMaxScaler standardization, and Gramian Angular Summation Field (GASF) image transformations.

4.2 Key Findings

During the first half of the project lifecycle, the primary engineering effort validated two core hypotheses:

- **Data Transformation Efficacy:** Transforming 1D tabular clinical records into 2D Gramian Angular Summation Fields (GASF) allowed the CNN baseline to effectively extract spatial relationships among distinct clinical features, proving the viability of our imaging-based approach for non-image data.
- **Architectural Decoupling:** The decision to build a state-driven, heavily validated frontend SPA independently of the backend ensured rapid UI iteration. The mock integration tests prove the frontend is fully capable of rendering the complex XAI heatmaps and narrative responses that the backend will eventually serve.

4.3 Future Work

Following our established timeline, our immediate next steps encompass expanding the deep learning model and establishing backend integrations:

1. **Advanced Model Training:** Expand the baseline CNN by integrating LSTM and Attention mechanisms to evaluate sequence patterns.
2. **Explainability Engine:** Implement the core Python logic for extracting Grad-CAM heatmaps, Attention weights, and SHAP values.
3. **System Integration:** Deploy the Flask backend to establish the RESTful bridge to our completed frontend SPA.

4.4 Final Remarks

The Cardio AI project remains on schedule and structurally sound. By prioritizing a transparent, highly interactive user interface alongside a solid deep learning foundation, we are well-positioned to deliver an Explainable AI diagnostic tool that bridges the gap between complex neural network predictions and practical clinical trust.